Chapter 1

Introduction

1.1 Underlying concepts

Spectroscopy is the study of absorption and emission of electromagnetic radiation by matter. Its goal is to measure the structural and dynamical properties of atoms, molecules and condensed matter with atomic to sub-atomic resolution. Over the past two centuries, spectroscopy has greatly contributed to our knowledge of the atomic-scale details of matter. Spectroscopy has revealed the equilibrium geometry of most small molecules, including chemical reaction intermediates and unstable species, such as radicals. Gerhard Herzberg (1904-1999) has been one of the leading figures in this area of science and was awarded the Nobel Prize in 1971. One representative achievement was the determination of the structure of methylene (CH_2) from the study of its absorption spectrum. The work of Herzberg is summarized in a three-volume book that will serve as the main reference for this lecture (see Appendix A and the website www.atto.ethz.ch/education/spectroscopy).

Long before the work of Herzberg, spectroscopy has had an even deeper impact on modern science. It has driven the development of quantum mechanics. The measurement of the emission spectrum of heated objects (so-called "black-body radiation") has led Max Planck (1858-1947) to his original quantum theory of radiation that we will discuss in detail below. This theory established the quantum nature of light, which is nowadays of greatest importance in areas as diverse as quantum information science and nano-photonics. Early measurements of atomic transitions lead to the breakthrough of the atomic model proposed by Niels Bohr (1885-1962) which initiated the development of quantum mechanics. Spectroscopic measurements also played a role in the discovery of the photoelectric effect by Albert

Einstein (1879-1955) which provided the foundations for photoelectron spectroscopy. The latter has been developed into a remarkably powerful tool to determine the nature and energetic sequence of molecular orbitals and to identify the elemental composition of matter (X-ray photoelectron spectroscopy).

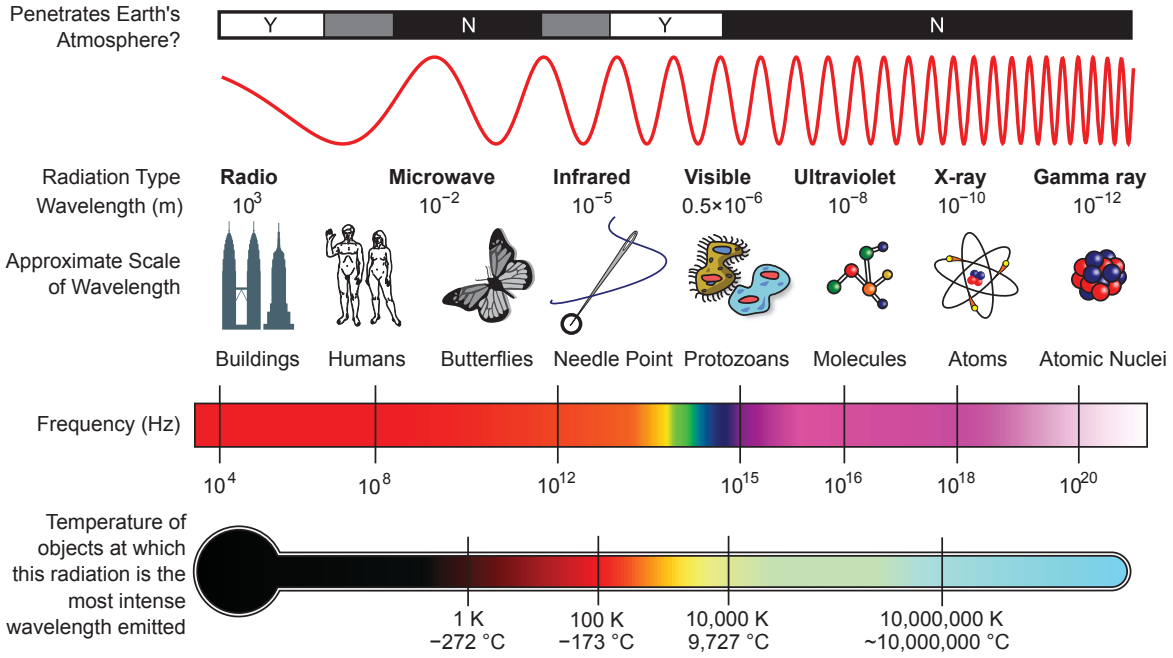


Figure 1.1: Illustration of the electromagnetic spectrum (taken from en.wikipedia.org/wiki/Electromagnetic_spectrum)

Analytical chemistry is an important sub-field that relies almost entirely on spectroscopy. As a reminder of previous lectures, we start with the study of the electromagnetic spectrum of light (Figure 1.1). Spectroscopy in the visible and ultraviolet regions probes transitions between different electronic states in atoms or molecules. This will be the main focus of the present lecture. Functional groups in molecules often possess electronic transitions at characteristic wavelengths, *e. g.* around 280 nm in the case of carbonyl (–C=O) groups, that allow their identification from spectral data. Transitions in the infrared region of the electromagnetic spectrum correspond to vibrational excitations in molecules which also have a great importance in analytical applications. Radiation in the microwave region drives transitions between rotational energy levels of molecules which provide accurate information on the equilibrium structure of molecules in the gas phase. Transitions in the radiofrequency range form the basis of magnetic resonance methods like nuclear magnetic resonance (NMR) and electron spin resonance (ESR) that have been studied in detail in the lecture Physical

Chemistry IV.

In the following sections of this first chapter, we will introduce the basic concepts and principles that underlie the whole field of spectroscopy. As is often the case in science, the major part of a field can be cast in a few leading concepts from which all the important knowledge can be derived.

1.2 Thermal radiation and Planck's law

We consider a cubic cavity of side length L held in equilibrium at a temperature T. The walls of the cavity emit and absorb radiation because of their thermal excitation. We would like to know the total energy density as well as the frequency distribution of the radiation inside the cavity.

Since the cavity is in thermal equilibrium, the emitted power $P_{\mathrm{e}}(\nu)$ has to be equal to the absorbed power $P_{\mathrm{a}}(\nu)$ for all frequencies ν. Inside the cavity, the radiation field is thus stationary and can be written as a superposition of plane waves

$$E(\vec{r},t) = \sum_p A_p \exp\left[\mathrm{i}(\omega_p t - \vec{k_p}\cdot\vec{r})\right] + \text{compl. conj.} \tag{1.1}$$

The electric field strength has to be zero on the walls of the cavity which imposes the boundary condition

$$E(x=0,y,z,t) = E(x=L,y,z,t) = 0 \quad \text{with } 0 \le y \le L \quad \text{and} \quad 0 \le z \le L \tag{1.2}$$

and similar conditions for the y and z dimensions. Consequently, the electric field takes the form

$$E(\vec{r},t) = A\sin(k_x x)\sin(k_y y)\sin(k_z z), \tag{1.3}$$

which imposes the boundary conditions on the wave vector

$$k_x = \frac{\pi}{L}n_1, k_y = \frac{\pi}{L}n_2, k_z = \frac{\pi}{L}n_3, \tag{1.4}$$

i. e. the cavity length must be an integer multiple of the half-wavelength $L = n_i\frac{\lambda_i}{2}$, a situation that is illustrated in Figure 1.2 in one dimension.

The wave vector of the radiation field can thus be written as

$$\vec{k} = \frac{\pi}{L}(n_1, n_2, n_3) \tag{1.5}$$

with positive integers n_i ($n_i \in \mathbb{N}^*$). The magnitude of the allowed wave vectors is thus

$$\left|\vec{k}\right| = \frac{\pi}{L}\sqrt{n_1^2 + n_2^2 + n_3^2} \tag{1.6}$$

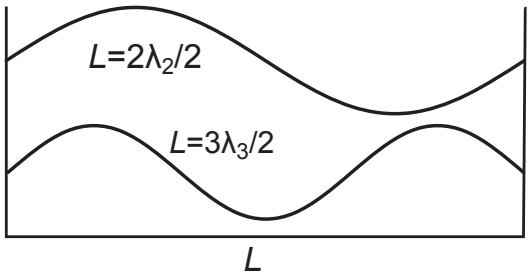


Figure 1.2: Allowed electro-magnetic waves in a cavity of length L.

which can be written in terms of the wavelength $\lambda = 2\pi/\left|\vec{k}\right|$, the frequency $\nu = c\left|\vec{k}\right|/2\pi$ or the angular frequency $\omega = c\left|\vec{k}\right|$.

The modes with angular frequency between 0 and ω fulfils

$$n_1^2 + n_2^2 + n_3^2 \leq \frac{\omega^2 L^2}{c^2\pi^2}. \tag{1.7}$$

In a Cartesian system with the coordinates $(\pi/L)(n_1, n_2, n_3)$ (see Figure 1.3), each triple (n_1, n_2, n_3) represents a point in a three-dimensional structure. If the radius is large compared to π/L, *i. e.* the wavelength is much smaller than the dimensions L of the cavity, Equation (1.7) represents the interior of a sphere with the radius

$$r = \sqrt{n_1^2 + n_2^2 + n_3^2} = \frac{\omega L}{c\pi}. \tag{1.8}$$

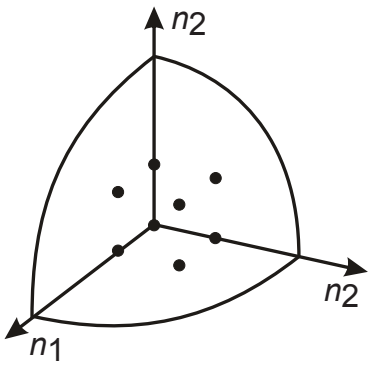


Figure 1.3: Diagram of the possible (n_1, n_2, n_3) combinations for occupation numbers of the modes of an optical cavity. The possible combinations are contained within an octant of a sphere of radius $r = \frac{\omega L}{c\pi}$.

In this limit, the number of modes can be estimated from the volume of the octant of the sphere shown in Figure 1.3. This volume is

$$V_{\text{oct}} = \frac{1}{8}\frac{4\pi}{3}\left(\frac{\omega L}{c\pi}\right)^3. \tag{1.9}$$

Since two polarization states are possible for each mode, this leads to the mode number

$$N_\omega(\omega) = 2V_{\text{oct}} = \frac{\pi}{3}\left(\frac{\omega L}{c\pi}\right)^3 \tag{1.10}$$

and, expressed in frequencies $\nu = \frac{\omega}{2\pi}$,

$$N(\nu) = \frac{8\pi\nu^3 L^3}{3c^3}. \tag{1.11}$$

In the following sections, we will often refer to the density of modes per unit volume within a frequency interval $\mathrm{d}\omega$ or $\mathrm{d}\nu$, which is obtained by differentiating the number of modes per volume $N_\omega(\omega)/V$ (here $V = L^3$ is the volume of the cavity) with respect to ω (or with respect to ν when expressed in frequencies)

$$g_\omega(\omega) = \frac{1}{V}\frac{\mathrm{d}N_\omega(\omega)}{\mathrm{d}\omega} = \frac{\omega^2}{c^3\pi^2}, \tag{1.12}$$

$$g(\nu) = \frac{1}{V}\frac{\mathrm{d}N(\nu)}{\mathrm{d}\nu} = \frac{8\pi\nu^2}{c^3}. \tag{1.13}$$

Each mode of the cavity described in the last setion is in equilibrium with the matter of the cavity walls. This matter can be described as classical mechanical oscillators in thermodynamic equilibrium at temperature T with a total mean energy of kT (see lecture Physical Chemistry II), where k is the Boltzmann constant ($k = 1.3806488(13) \times 10^{-23}\,\mathrm{J\,K^{-1}}$). This implies that the mean energy in each mode is also kT and thus the energy content in the frequency interval ν to ν+$\mathrm{d}\nu$ of a radiation field in a volume V at a temperature T is given by

$$U(\nu)\mathrm{d}\nu = g(\nu)VkT\mathrm{d}\nu = \frac{8\pi\nu^2}{c^3}VkT\mathrm{d}\nu. \tag{1.14}$$

and the energy density is given by

$$\rho(\nu) = \frac{U(\nu)}{V} = g(\nu)kT = \frac{8\pi\nu^2}{c^3}kT. \tag{1.15}$$

This relation is also known as the **Rayleigh-Jeans law** and correctly predicts the radiation density for low frequencies ($h\nu \ll kT$) and long wavelengths, *i. e.* in the infrared region. However, it strongly disagrees with experimental observations in the high-frequency ($h\nu \gg kT$) or short wavelength region because it predicts that the energy density becomes infinite. This constituted a major puzzle in the late 1800's and was called the "UV catastrophy".

To resolve this discrepancy, Max Planck proposed that each mode of the radiation field can only emit or absorb energy in integer multiples q of a fundamental energy quantum $h\nu$, where h is Planck's constant ($h = 6.62606957(29) \times 10^{-34}\,\mathrm{J\,s}$). This quantum is nowadays called the

"photon" and each mode of the cavity can thus only contain an integer number of photons and thus carry the energy $qh\nu$.

The energy E (in J, usually in eV) carried by one photon can be expressed as a function of its frequency ν (in Hz, *i. e.* s^{-1}), its circular frequency ω (in s^{-1}), its wavelength λ (in m, usually nm) or its wavenumber $\tilde{\nu}$ (in m^{-1}, usually cm^{-1}) as

$$E = h\nu = \hbar\omega = \frac{hc}{\lambda} = hc\tilde{\nu}. \tag{1.16}$$

It is very useful to remember *approximate* conversion factors between these units which are as follows

$$1.000\,\text{eV} \leftrightarrow 2.418 \times 10^{14}\,\text{Hz} \leftrightarrow 8066\,\text{cm}^{-1}. \tag{1.17}$$

The partitioning of energy into the different possible modes in thermal equilibrium is governed by the Maxwell-Boltzmann distribution stating that the probability $p(q)$ to find the energy $qh\nu$ in a specific mode is

$$p(q) = \frac{1}{Z}\exp\left(-\frac{qh\nu}{kT}\right), \tag{1.18}$$

where

$$Z = \sum_q \exp\left(-\frac{qh\nu}{kT}\right) \tag{1.19}$$

is the so-called partition function (see Lecture Physical Chemistry VI - Statistical thermodynamics). The role of Z is to ensure that the sum of all probabilities is one, *i. e.* $\sum_q p(q) = 1$. One can thus define the mean energy per mode as

$$\langle E(\nu)\rangle = \sum_{q=0}^{\infty} p(q)qh\nu = \frac{\sum_q qh\nu \exp\left(-qh\nu/(kT)\right)}{\sum_q \exp\left(-qh\nu/(kT)\right)}, \tag{1.20}$$

which leads to

$$\langle E(\nu)\rangle = \frac{h\nu}{\exp\left(h\nu/(kT)\right) - 1}. \tag{1.21}$$

The energy density of the radiation field within the frequency interval ν to $\nu + \mathrm{d}\nu$ is given by the product of the mode density $g(\nu)$ (Equation (1.13)) per unit volume and the mean energy $\langle E(\nu)\rangle$ per mode, which constitutes **Planck's law**

$$\rho(\nu)\mathrm{d}\nu = g(\nu)\langle E(\nu)\rangle\mathrm{d}\nu = \frac{8\pi\nu^2}{c^3}\frac{h\nu}{\exp\left(h\nu/(kT)\right) - 1}\mathrm{d}\nu. \tag{1.22}$$

This relation characterizes the spectral energy distribution of a radiation field that is in thermal equilibrium with its environment at a temperature T. Equation (1.22) is represented graphically in Figure 1.4 for the temperatures T=293 K, 2500 K and 3500 K. This graph

shows why a solid metallic body at room temperature does not emit any visible radiation, whereas the same body emits red light when heated to 2500 K and eventually the whole visible spectrum when heated to 3500 K, which makes it look white. Other examples of radiation sources that closely follow Planck's law are the sun, the tungsten wire of a light bulb and flash lamps.

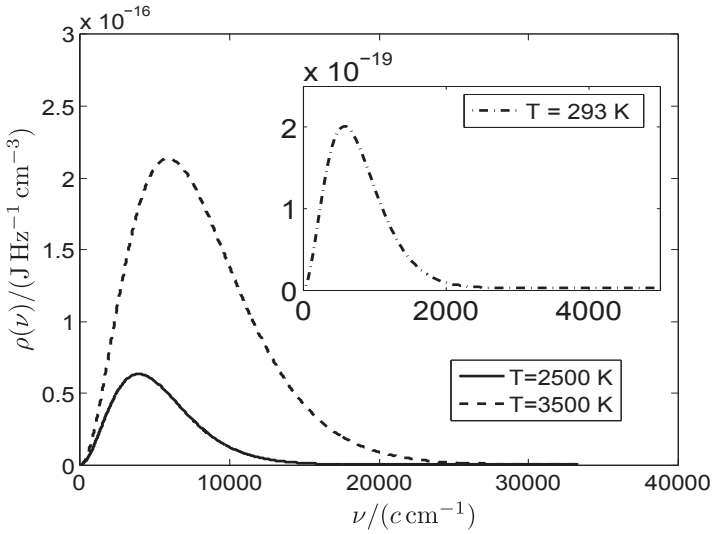


Figure 1.4: Spectral distribution of the energy density $\rho(\nu)$ for different temperatures (Planck's law).

Table 1.1 lists the frequencies and wavenumbers corresponding to the maximum of the energy density $\rho(\nu)$ for different temperatures T. The cosmic background radiation (T=2.7 K) has its maximum in the far-infrared. The thermal emission of the earth's surface peaks in the mid-infrared, close to the bending frequency of CO_2 (667 cm^{-1}), explaining why this molecule is the main carrier of global warming. The maximum of the sun's emission spectrum lies close to the visible range.

T/K	ν_{max}/Hz	$\tilde{\nu}_{max}$/cm^{-1}	$\lambda(\nu_{max})/\mu$m	region
3	1.76×10^{11}	5.88	1700	cosmic background
300	1.76×10^{13}	588	17	surface of the earth
6000	3.53×10^{14}	11768	0.85	surface of the sun

Table 1.1: Frequencies ν_{max}, wavenumbers $\tilde{\nu}_{max}$ and wavelengths $\lambda(\nu_{max})$ describing the maximum of the energy density distribution $\rho(\nu)$ for different temperatures T.

Other sources of radiation, like gas-discharge lamps (neon tubes) and lasers do not follow

Planck's law and are also known as "non-thermal" radiation sources.

1.3 Absorption and emission of radiation

This section discusses the absorption and emission of thermal radiation by a two-level system in the framework of a kinetic model, introduced by A. Einstein (A. Einstein, *Physik Z.* **18**, 121 (1917)). We consider a system with two states, labelled 1 and 2, of energies E_1 and E_2 with $E_1 < E_2$, in contact with thermal radiation. The system is initially in the lower state 1. The possible processes resulting from the interaction of the system with radiation are illustrated in Figure 1.5. The system can be excited to state 2 by **absorption** of radiation at frequency

$$\nu_{12} = \frac{E_2 - E_1}{h}, \tag{1.23}$$

a transition that will be designated with the notation $2 \leftarrow 1$. In a kinetic formalism, one can thus write down the rates of change of the populations N_i in the two states as

$$-\frac{\mathrm{d}N_1^{(\mathrm{abs})}}{\mathrm{d}t} = \frac{\mathrm{d}N_2^{(\mathrm{abs})}}{\mathrm{d}t} = B_{12}\rho(\nu_{12})N_1, \tag{1.24}$$

where $\rho(\nu_{12})$ is the photon density and B_{12} depends on the properties of the system, in particular the wave functions of the states 1 and 2, but not on the temperature.

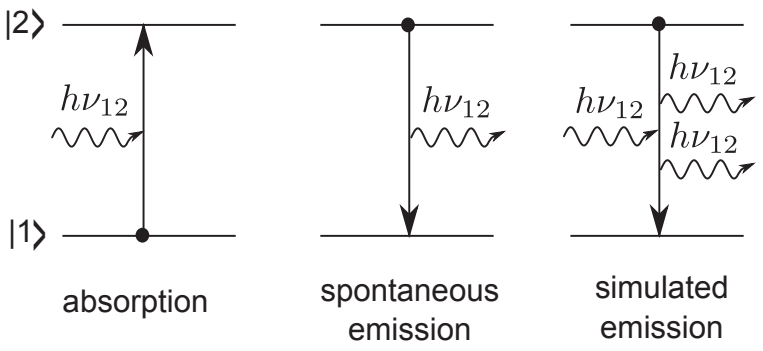


Figure 1.5: Interaction of a two-level system with a radiation field. The black dot labels the state of the system prior to the radiative process.

The excited system can spontaneously decay from state 2 to state 1 under emission of light at the same frequency ν_{12} defined above. This process is written $2 \rightarrow 1$ and can again be described kinetically

$$\frac{\mathrm{d}N_1^{(\mathrm{sp.\,em.})}}{\mathrm{d}t} = -\frac{\mathrm{d}N_2^{(\mathrm{sp.\,em.})}}{\mathrm{d}t} = A_{21}N_2. \tag{1.25}$$

This process is called **spontaneous emission** because it is independent of the radiation field. A_{21} depends on the properties of the system but is temperature-independent. Spontaneous emission is characterized by an arbitrary direction of emission.

Finally, the radiation field can also **induce** a transition from state 2 to state 1,

$$\frac{\mathrm{d}N_1^{(\text{stim. em.})}}{\mathrm{d}t} = -\frac{\mathrm{d}N_2^{(\text{stim. em.})}}{\mathrm{d}t} = B_{21}\rho(\nu_{12})N_2. \tag{1.26}$$

This process is called **stimulated emission**. An important property of stimulated emission is that it occurs along the direction imposed be the incoming radiation. In the following paragraphs, the relation between the three coefficients B_{12}, B_{21} and A_{21} will be derived.

We now consider the system with the two states, 1 and 2, to be in equilibrium with a thermal radiation field. Once the thermal equilibrium is established, the rate of absorption and the total rate of emission (spontaneous and stimulated) must balance each other, a situation that is described by the following equation:

$$\left(\frac{\mathrm{d}N_1}{\mathrm{d}t}\right)_{\text{eq}} = -B_{12}\rho(\nu_{12})N_1^{(\text{eq})} + A_{21}N_2^{(\text{eq})} + B_{21}\rho(\nu_{12})N_2^{(\text{eq})} = 0 \tag{1.27}$$

or

$$B_{12}\rho(\nu_{12})N_1^{(\text{eq})} = \left[A_{21} + B_{21}\rho(\nu_{12})\right] N_2^{(\text{eq})}. \tag{1.28}$$

Since the system is simultaneously in thermal equilibrium with the environment at temperature T, the ratio of populations $N_1^{(\text{eq})}$ and $N_2^{(\text{eq})}$ has to fulfill

$$\frac{N_2^{(\text{eq})}}{N_1^{(\text{eq})}} = \exp\left(-\frac{E_2 - E_1}{kT}\right) = \exp\left(-\frac{h\nu_{12}}{kT}\right), \tag{1.29}$$

giving

$$\rho(\nu_{12}) = \frac{A_{21}/B_{21}}{\left(\frac{B_{12}}{B_{21}}\right)\exp\left(\frac{h\nu_{12}}{kT}\right) - 1}. \tag{1.30}$$

Since Equations (1.30) and (1.22) must be simultaneously fulfilled at the same temperature, equating the coefficients in the two equations leads to

$$B_{12} = B_{21} \quad \text{and} \quad A_{21} = \frac{8\pi h\nu_{12}^3}{c^3}B_{21}. \tag{1.31}$$

These two relations have many important fundamental implications. First, the Einstein coefficients for absorption and stimulated emission are equal. This result will become obvious in the next paragraph and in later chapters where we will calculate these quantities from first principles. Second, the rate of spontaneous emission scales with ν_{12}^3. This means that the lifetime of excited states is much longer in the case of pure vibrational excitation (typically

milliseconds) than in the case of pure electronic excitation (typically nanoseconds). This scaling law also explains why it is much more difficult to build lasers in the ultraviolet to X-ray wavelength range than in the visible or infrared range. The rate of spontaneous emission is so large that it becomes difficult or impossible to maintain a population inversion as will be discussed in the next sections.

Since all three Einstein coefficients are temperature-independent, one can choose one of them (*e. g.* B_{12}) to express the others. In later sections, we will show that B_{12} can be expressed within the dipole approximation, in terms of the wave functions of the two states

$$B_{12} = \frac{8\pi^3}{3h^2(4\pi\varepsilon_0)} \left| \langle 2|\hat{\vec{\mu}}|1\rangle \right|^2, \tag{1.32}$$

where $|1\rangle = \Psi_1$ represents the wave function of state 1, $\langle 2| = \Psi_2^*$ represents the complex conjugate of the wave function of state 2, $\hat{\vec{\mu}}$ is the dipole operator and ε_0 is the vacuum permeability ($\varepsilon_0 = 8.854187817 \times 10^{-12}\,\mathrm{F\,m^{-1}}$).

Finally, the relation (1.30) can also be used to calculate the average number of photons that occupy each mode of the electromagnetic field at a certain temperature T. Each mode can only be occupied by an integer number of photons, but the average number of photons per mode is given by:

$$n(\nu, T) = \frac{\langle E(\nu)\rangle}{h\nu} = \frac{1}{\exp\left(h\nu/(kT)\right) - 1}. \tag{1.33}$$

At room temperature, in the visible range (λ=600 nm), one finds $\exp(h\nu_{\mathrm{vis}}/(kT_{\mathrm{amb}})) \approx 4 \times 10^{35}$ and $n(\nu_{\mathrm{vis}}, T_{\mathrm{amb}}) \approx 2.5 \times 10^{-36}$. Consequently, the spontaneous emission per mode exceeds by far the thermally stimulated emission. However, it is possible to concentrate the radiation energy into a small number of modes, *e. g.* in a laser, in which case the occupation number $n(\nu, T)$ can become much larger than one. In the high-temperature limit $\exp\left(h\nu/(kT)\right) \to 1$ and the number of photons per mode can also become very large.

1.4 Measurable quantities in spectroscopy

The goal of this section is to establish a link between the quantities that are typically measured in frequency-resolved experiments and the Einstein coefficients. The simplest spectroscopic experiment consists in monitoring the absorption of a monochromatic beam of light by a sample of atoms or molecules as illustrated in Figure 1.6. The particles (atoms or molecules), at density C are initially all in the lower state, labelled "1", and are contained in a transparent

glass cell. They can absorb the radiation of frequency ν_{12} and undergo a transition to an upper state labelled "2".

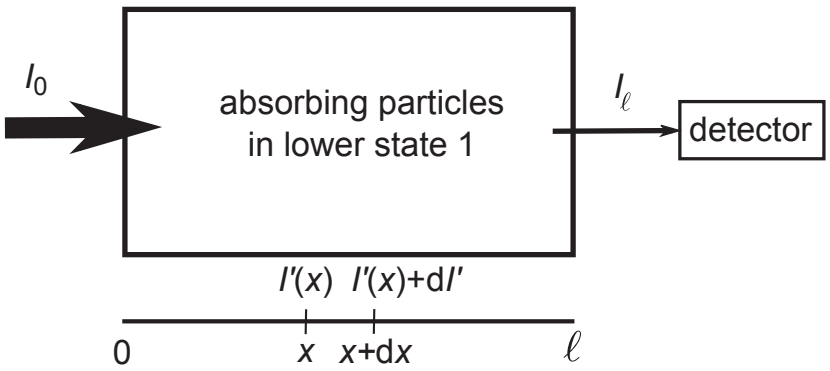


Figure 1.6: Experimental scheme of an absorption measurement

The intensity I of the beam of light can be expressed as the flow of photons per unit surface and unit time

$$I = cC_\gamma, \tag{1.34}$$

where c is the speed of light and C_γ is the photon concentration. We note that this differs from the traditional definition of intensity I_E which is however recovered by multiplying I with the photon energy

$$I_E = Ih\nu. \tag{1.35}$$

Absorption of light across the cell of length ℓ can be expressed through the differential decrease of the intensity $\mathrm{d}I$ over the distance $\mathrm{d}x$. $\mathrm{d}I$ is proportional to the intensity at position x, the concentration of molecules C and the length $\mathrm{d}x$

$$-\,\mathrm{d}I'(x) = \sigma_{12} I'(x) C \mathrm{d}x, \tag{1.36}$$

where σ_{12} has the dimension of a surface and is called the **absorption cross section**. At the position of the detector the measured intensity is

$$-\int_{I_0}^{I_\ell} \frac{\mathrm{d}I'}{I'} = \int_0^\ell \sigma_{12} C \mathrm{d}x, \tag{1.37}$$

which leads to the **Lambert-Beer law**

$$\ln\left(\frac{I_0}{I_\ell}\right) = \sigma_{12} C \ell. \tag{1.38}$$

Another form of this law is

$$A_{10} = \log_{10}\left(\frac{I_0}{I_\ell}\right) = \varepsilon_{12} C_{\mathrm{mol}} \ell, \tag{1.39}$$

where A_{10} is the *decadic* absorbance, ε_{12} is the molar extinction coefficient in $\mathrm{m^2\,mol^{-1}}$ and C_{mol} is the molar concentration in $\mathrm{mol\,m^{-3}}$.

Absorption can also be described as a bimolecular reaction between particles of concentration C in state i and photons γ. One can define a bimolecular rate coefficient for the decrease of the photon concentration C_γ

$$-\frac{\mathrm{d}C_\gamma}{\mathrm{d}t} = k_{\mathrm{bi}} C_\gamma C. \tag{1.40}$$

Using the photon speed c, one can transform from position to time

$$c = \frac{\mathrm{d}x}{\mathrm{d}t} \tag{1.41}$$

$$-\frac{\mathrm{d}I(x)}{\mathrm{d}x} = -\frac{\mathrm{d}C_\gamma c}{\mathrm{d}x} = -\frac{\mathrm{d}C_\gamma}{\mathrm{d}x}\frac{\mathrm{d}x}{\mathrm{d}t} = -\frac{\mathrm{d}C_\gamma}{\mathrm{d}t}. \tag{1.42}$$

The absorbing species can be considered to be at rest relative to the photons, which gives

$$-\frac{\mathrm{d}C_\gamma}{\mathrm{d}t} = k_{\mathrm{bi}} C_\gamma C = \sigma_{12} c C_\gamma C. \tag{1.43}$$

The bimolecular rate constant is thus

$$k_{\mathrm{bi}} = \sigma_{12} c, \tag{1.44}$$

which leads to the definition of a pseudo first-order rate constant (assuming that C_γ is constant)

$$k_{\mathrm{first}} = \sigma_{12} c C_\gamma. \tag{1.45}$$

At this point, we would like to compare k_{first} with the pseudo-first order rate coefficients defined on the basis of the Einstein coefficients $K_{12} = B_{12}\rho(\nu_{12})$ and $K_{21} = B_{21}\rho(\nu_{12})$. Just as in the case of a damped classical oscillator, a transition between two quantum mechanical states has a *finite* linewidth that results from the lifetime of the excited state. An experimentally measured line is subject to additional broadening mechanisms, like Doppler broadening and power broadening that will be discussed in exercises or in later sections. This requires the consideration of the photon density (or concentration) in the frequency interval between ν and $\nu + \mathrm{d}\nu$

$$\mathrm{d}C'_\gamma = C'_\gamma(\nu)\mathrm{d}\nu \tag{1.46}$$

with

$$C_\gamma = \int_0^\infty C'_\gamma(\nu)\mathrm{d}\nu. \tag{1.47}$$

Considering their spectral dependence, the rate constants for absorption K_{12} and for stimulated emission K_{21} introduced previously are obtained by integrating over all frequencies contributing to the transition $2 \leftrightarrow 1$

$$K_{12} = \int_0^\infty K_{12}(\nu)\mathrm{d}\nu = \int_0^\infty c\sigma_{12}(\nu)C'_\gamma(\nu)\mathrm{d}\nu. \tag{1.48}$$

We note that $\sigma_{12}(\nu)$ depends on the frequency ν (it peaks strongly at the resonance frequency), whereas $C'_\gamma(\nu)$ depends on the light source. Therefore, it is sufficient to evaluate the integrals in the vicinity of the resonance frequency, which is expressed by the substitution

$$\int_0^\infty \rightarrow \int_{\text{line}} . \tag{1.49}$$

$C'_\gamma(\nu)$ is related to the energy density $\rho(\nu)$ through $C'_\gamma(\nu) = \frac{\rho(\nu)}{h\nu}$ and therefore,

$$K_{12} = \frac{c}{h}\int_{\text{line}} \frac{\sigma_{12}(\nu)}{\nu}\rho(\nu)\mathrm{d}\nu. \tag{1.50}$$

When the photon density of the excitation source varies little with its frequency (so-called "white-light" excitation), $\rho(\nu) \approx \rho(\nu_{12})$ and

$$K_{12} \approx \frac{c}{h}\rho(\nu_{12})\int_{\text{line}} \frac{\sigma_{12}(\nu)}{\nu}\mathrm{d}\nu. \tag{1.51}$$

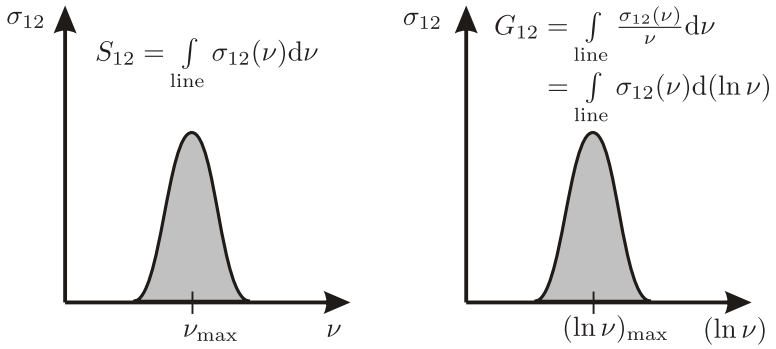


Figure 1.7: Line strength S_{12} and integrated absorption cross section G_{12} of an absorption line centered at $\nu_{\max}$.

With the definition of the **integrated absorption cross section** G_{12} (see Figure 1.7)

$$G_{12} = \int_{\text{line}} \frac{\sigma_{12}(\nu)}{\nu}\mathrm{d}\nu, \tag{1.52}$$

one obtains

$$\begin{aligned} K_{12} &\approx \frac{c}{h}\rho(\nu_{12})G_{12} \\ \text{and } B_{12} &\approx \frac{c}{h}G_{12}. \end{aligned} \tag{1.53}$$

If the photon density of the light source varies strongly with its frequency, *i. e.* $\rho(\nu)$ is not approximately constant, K_{12} must be calculated using Equation (1.50).

The **line strength** is defined as

$$S_{12} = \int_{\text{line}} \sigma_{12}(\nu) \mathrm{d}\nu, \tag{1.54}$$

as illustrated in Figure 1.7. Using this definition, the integrated absorption cross section can be approximated as

$$G_{12} \approx \frac{1}{\nu_{12}^{\max}} S_{12} \tag{1.55}$$

and

$$G_{12} \approx \frac{c^2}{8\pi\nu_{12}^3} A_{21}. \tag{1.56}$$

We recall that the Einstein coefficient B_{12} can be expressed as a function of the transition moment $M_{12} = \langle 2|\hat{\vec{\mu}}|1\rangle$

$$B_{12} = \frac{8\pi^3}{3h^2(4\pi\varepsilon_0)} |M_{12}|^2 \tag{1.57}$$

and for the integrated absorption cross section

$$G_{12} \approx \frac{8\pi^3}{3hc(4\pi\varepsilon_0)} |M_{12}|^2 . \tag{1.58}$$

1.5 Population inversion and laser operation

A beam of light of frequency ν_{12} incident on a sample of atoms or molecules in the excited state 2 (*i. e.* $C_1 = 0$) will be *amplified* according to the Lambert-Beer law

$$I = I_0 \exp\left(\sigma_{12} C_2 \ell\right). \tag{1.59}$$

More generally, light will undergo **net absorption** if $C_1 > C_2$ but **net amplification** if $C_1 < C_2$. The net gain can be expressed as

$$\frac{I}{I_0} = \exp\left[\sigma_{12}\left(C_2 - C_1\right)\ell\right]. \tag{1.60}$$

The situation $C_1 < C_2$ is called **population inversion** and underlies the operation of the laser. LASER stands for Light Amplification by Stimulated Emission of Radiation. Historically, the laser was preceded by the maser, an amplification device operating in the microwave region (hence maser), based on population inversion in ammonia molecules.

Amplification of light alone is not sufficient for the construction of a laser, but one also needs a feedback structure that will make light pass through the gain medium several times, as

illustrated in Figure 1.8. This is most commonly realized using total reflection on a mirror on one end of the cavity and a partially reflective mirror on the other end. The major part of the radiation is reflected back into the cavity, where it is amplified further, whereas a small part is allowed to escape. This fraction forms the laser beam that is characterized by a very low divergence and a high degree of temporal and spatial coherence.

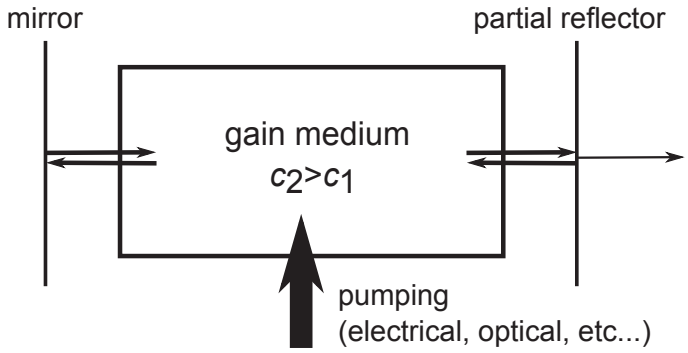


Figure 1.8: Experimental setup of a laser

Many different materials and processes can be used in the construction of a laser. Based on the operation principle, they can be grouped into three categories illustrated in Figure 1.9. The two-level laser relies exclusively on the principles of absorption and stimulated emission discussed in the last sections. An example of this type of device is the original ammonia maser. The problem with this type of laser is that it tends to turn itself off because the amplification is suppressed when strong stimulated emission occurs. In the ammonia maser, this problem is solved by pumping away the ground state molecules, but this approach is limited to gas lasers. In general, a two-level laser is thus very unlikely to work. The 3-level laser avoids this problem by adding one step of radiationless transition into the cycle. The excitation process which is also called "pumping" is spectrally distinct from the emitted radiation. The radiationless transition, shown as a wavy line, can either serve to populate the upper level or to depopulate the lower level of the laser transition. In both cases, strong stimulated emission will not counteract the effect of pumping which allows the laser to run with much higher yield. An example of the 3-level laser is the ruby laser, the first ever demonstrated by Theodore Maiman in 1960. Pumping is achieved by optical excitation of the ruby crystal with flashlamps and lasing occurs at 694.3 nm. Finally, in the 4-level laser, the lasing transition is decoupled form the pump transition through two steps of non-radiative transitions. Examples of this kind include most practically useful lasers, including the high-power CO_2 lasers and the titanium-sapphire (Ti:Sa) lasers. The CO_2 lasers are pumped by high-voltage

discharge, whereas Ti:Sa lasers are optically pumped by another laser.

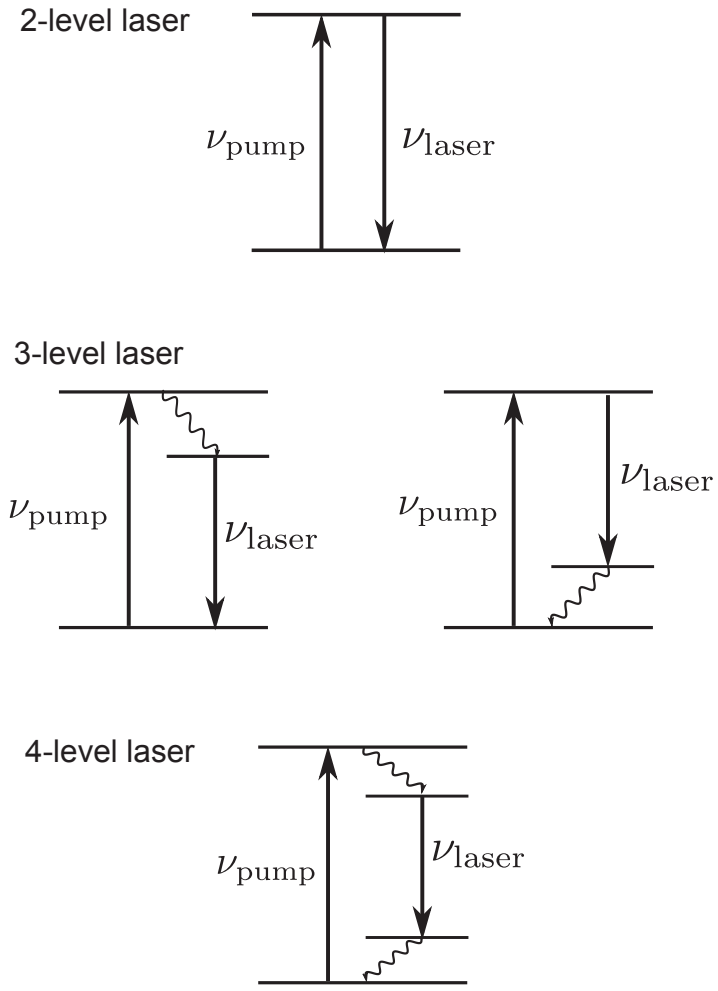


Figure 1.9: Operation principles of the three most common types of lasers

The following paragraphs will present a derivation of the equations underlying laser operation. In the preceding sections, we have calculated the occupation numbers of radiation modes in thermal equilibrium. We will now calculate these numbers in the non-equilibrium conditions associated with population inversion. The rate of photon production in an atomic or molecular sample with population numbers N_1 and N_2 in states 1 and 2 is

$$\frac{\mathrm{d}N_\gamma}{\mathrm{d}t} = (N_2 - N_1)\, B_{21}\rho'(\nu_{12}) + A_{21}N_2, \tag{1.61}$$

where $\rho'(\nu_{12})$ is the radiation density. Photons are produced within a bandwidth $\Delta\nu$ around the resonance frequency ν_{12}

$$N_\gamma = g(\nu_{12})\Delta\nu n(\nu_{12}), \tag{1.62}$$

where $g(\nu_{12})$ is the mode density as defined in Equation (1.13) and $n(\nu_{12})$ is the number of photons per mode. The total number of photons N_γ and $n(\nu)$ are time-dependent, whereas $g(\nu)$ and $\Delta\nu$ are time-independent. Using Equation (1.61) one thus obtains

$$\frac{\mathrm{d}n(\nu_{12})}{\mathrm{d}t} = (N_2 - N_1)\,\frac{B_{21}\rho'(\nu_{12})}{g(\nu_{12})\Delta\nu} + \frac{A_{21}}{g(\nu_{12})\Delta\nu}N_2. \tag{1.63}$$

Since $V\rho'(\nu_{12}) = g(\nu_{12})h\nu_{12}n(\nu_{12})$, using $g(\nu_{12})$ from Equation (1.13), it follows that

$$V\rho'(\nu_{12}) = \frac{8\pi V}{c^3}\nu_{12}^2 h\nu_{12}n(\nu_{12}) \tag{1.64}$$

and with the expressions for A_{21} and B_{21} from Equation (1.31), we obtain

$$B_{21}\rho'(\nu_{12}) = \frac{A_{21}c^3}{8\pi h\nu_{12}^3}\rho'(\nu_{12}) = A_{21}n(\nu_{12}). \tag{1.65}$$

With the definition

$$W = \frac{A_{21}}{g(\nu_{12})\Delta\nu}, \tag{1.66}$$

we obtain the **laser equation**

$$\frac{\mathrm{d}n(\nu_{12})}{\mathrm{d}t} = (N_2 - N_1)\,Wn(\nu_{12}, t) + WN_2, \tag{1.67}$$

where $(N_2 - N_1)W$ represents the **gain**. When the pump process is fast enough, one can assume that N_1 and N_2 reach a stationary regime, which allows one to solve the differential equation (1.67) with the boundary condition $n(\nu_{12}) = 0$ at $t = 0$:

$$n(\nu_{12}, t) = \frac{N_2}{N_2 - N_1}\left(\exp\left(W(N_2 - N_1)t\right) - 1\right). \tag{1.68}$$

In the following, we will analyze the regions of stability of the laser equation.

- if $N_2 < N_1$, $n(\nu_{12}, t \to \infty) = -\frac{N_2}{N_2 - N_1} > 0$
  The number of photons per mode is fixed, which corresponds to a stable condition. However, this situation does *not* corresponds to net amplification of light.

- if $N_2 \approx N_1$, a series expansion of the exponential term gives $\exp\left(W\left(N_2 - N_1\right)t\right) \approx 1 + W\left(N_2 - N_1\right)t + ...$, *i. e.* $n(\nu_{12}, t) \approx N_2 Wt$, corresponding to a linear growth of the photon population with time.

- if $N_2 > N_1$, $n(\nu_{12}, t \to \infty) = \frac{N_2}{N_2 - N_1}\exp\left(W\left(N_2 - N_1\right)t\right)$, the photon number diverges with time and even the fastest pump process can not maintain the population difference.

Losses from the cavity by, *e. g.* scattering or a partially transmitting mirror lead to an additional term in the laser equation

$$\frac{\mathrm{d}n(\nu_{12})}{\mathrm{d}t} = (N_2 - N_1)\, W n(\nu_{12}) + W N_2 - k_1 n(\nu_{12}), \tag{1.69}$$

where $t_1 = 1/k_1$ is the decay time of the laser resonator and $k_1 n(\nu_{12})$ represents the **losses**. For large values of $n(\nu_{12}, t)$ the contribution from spontaneous emission is often negligible, leading to the condition of laser operation

$$\frac{\mathrm{d}n(\nu_{12})}{\mathrm{d}t} > 0 \quad \text{or} \quad (N_2 - N_1)\, W > k_1, \tag{1.70}$$

i. e. the gain has to be larger than the losses, which provides the condition of operation for a laser

$$\frac{N_2 - N_1}{V} \frac{c^3}{8\pi\nu^2} \frac{A_{21}}{\Delta\nu} > k_1. \tag{1.71}$$

To summarize, laser operation needs population inversion and a gain larger than the losses.

1.6 Interaction of light and matter

An electromagnetic wave can be represented by a number of harmonic oscillators corresponding to the modes of the field. In the case of laser radiation, the number of quanta per field mode is extremely large, on the order of $10^{10} - 10^{20}$. In such cases, electromagnetic fields can be treated classically for all practical purposes and are well described by Maxwell's equations. In this lecture, atoms and molecules will be treated quantum mechanically, but the radiation field will be treated classically. The interaction of light with particles (atom or molecules) is greatly simplified when the wavelength of the radiation λ is much larger then the extension d of the particle

$$\lambda \gg d, \tag{1.72}$$

which is typically the case in the ultraviolet to infrared ($100\,\text{nm} - 100\,\mu\text{m}$) range of the electromagnetic spectrum. However, this is no longer true for X-rays and γ-rays.
The electrical dipole interaction of a monochromatic linearly polarized wave of circular frequency ω with a system of charged particles is

$$\hat{V}_{\text{el}}(t) = -\left(\sum_i q_i \vec{r}_i\right) \vec{E}_0^{(z)} \cos(\omega t + \phi), \tag{1.73}$$

where $\vec{E}_0^{(z)}$ is the peak electric field and $\vec{r}_i$ is the position of the particle i of charge q_i.
Writing the electric dipole moment $\vec{\mu}_{\text{el}} = \sum_i q_i \vec{r}_i$

$$\hat{V}_{\text{el}}(t) = -\vec{\mu}_{\text{el}} \vec{E}_0^{(z)} \cos(\omega t + \phi), \tag{1.74}$$

and by analogy for the interaction with the magnetic field

$$\hat{V}_{\text{mag}}(t) = -\vec{\mu}_{\text{mag}} \vec{B}_0^{(x)} \cos(\omega t + \phi). \tag{1.75}$$

In the optical domain, the electric dipole interaction dominates. However, important methods, like nuclear magnetic resonance (NMR) and electron paramagnetic resonance (EPR) rely on the magnetic dipole interaction (see lecture Physical Chemistry IV).

1.7 Coherent excitation of a two-level system

Many phenomena in nature are time-dependent and are described by the time-dependent Schrödinger equation that has been discussed in the lecture Physical Chemistry III.

REMINDER: The **time-dependent Schrödinger equation** of a n-particle system can be written as follows:

$$\mathrm{i}\hbar\frac{\partial\,\Psi(\vec{r}_1,\vec{r}_2,...,\vec{r}_n,t)}{\partial t} = \hat{H}\,\Psi(\vec{r}_1,\vec{r}_2,...,\vec{r}_n,t), \tag{1.76}$$

with coordinates $\vec{r}_j$ of particle j. The general solution of this equation is

$$\Psi(t) = \hat{U}(t,t_0)\,\Psi(t_0), \tag{1.77}$$

where $\Psi(t_0)$ describes the initial state of the system.

The **time-evolution operator** $\hat{U}$ is obtained from

$$\mathrm{i}\hbar\frac{\partial\hat{U}}{\partial t} = \hat{H}\hat{U}(t,t_0). \tag{1.78}$$

(i) Stationary (time-independent) states can be written in the form

$$\Psi(\vec{r},t) = \Psi(\vec{r})\exp\left(-\frac{\mathrm{i}Et}{\hbar}\right) \tag{1.79}$$

where E is the energy of the state described by $\Psi(\vec{r})$ and $\vec{r}$ is used collectively to represent all coordinates $(\vec{r}_1,\vec{r}_2,...,\vec{r}_n)$. Inserting Equation (1.79) into Equation (1.76) using the relation

$$\frac{\mathrm{d}}{\mathrm{d}t}\exp\left(-\frac{\mathrm{i}Et}{\hbar}\right) = -\frac{\mathrm{i}E}{\hbar}\exp\left(-\frac{\mathrm{i}Et}{\hbar}\right) \tag{1.80}$$

gives

$$E\,\Psi(\vec{r}) = \hat{H}\,\Psi(\vec{r}), \tag{1.81}$$

which is the **time-independent Schrödinger equation**.

Using the index k for the eigenfunction Ψ_k with energy E_k, one can thus write all stationary solutions of the time-dependent Schrödinger equation in the form

$$\Psi_k(\vec{r},t) = \Psi_k(\vec{r})\exp\left(-\frac{\mathrm{i}E_kt}{\hbar}\right). \tag{1.82}$$

The probability density associated with such a state is given by

$$\Psi_k(\vec{r},t)\,\Psi_k^*(\vec{r},t) = |\,\Psi_k(\vec{r},t)|^2 = |\,\Psi_k(\vec{r})|^2\,, \tag{1.83}$$

which is also time independent.

(ii) General time-dependent states: In this case, it is useful to remember the **superposition principle**. If $\Psi_1(\vec{r},t)$ and $\Psi_2(\vec{r},t)$ are two dynamical states of a system, then

$\Psi(\vec{r},t) = c_1\,\Psi_1 + c_2\,\Psi_2$ and in general, with complex c_k,

$$\Psi(\vec{r},t) = \sum_k c_k\,\Psi_k(\vec{r},t)\exp\left(-\frac{\mathrm{i}E_k t}{\hbar}\right) \tag{1.84}$$

are also possible dynamical states of the system. In this case $\Psi(\vec{r},t)\,\Psi^*(\vec{r},t)$ is time dependent. The probability of measuring the energy E_k is $c_k c_k^* = |c_k|^2$. In a time-dependent state, the energy itself is not well-defined, only its probability distribution.

In the following, we discuss the quantum mechanical description of the excitation of a N-level system. The coupling of a quantum mechanical system with radiation can be described by the time-dependent Schrödinger equation

$$\mathrm{i}\hbar\frac{\partial\,\Psi}{\partial t} = \hat{H}(t)\,\Psi, \tag{1.85}$$

where

$$\hat{H}(t) = \hat{H}_{\mathrm{mol}} - \vec{\mu}_{\mathrm{el}}\vec{E}_0^{(z)}\cos(\omega t + \phi). \tag{1.86}$$

Because $\hat{H}$ is time-dependent, the solution of the Schrödinger equation is often not available in closed form. It is often convenient to expand $\Psi(t)$ in eigenfunctions of the system, *e. g.* into molecular eigenfunctions

$$\hat{H}_{\mathrm{mol}}\phi_k = E_k\phi_k \tag{1.87}$$

i. e.

$$\Psi(t) = \sum_k b_k(t)\phi_k(\vec{r}_1,...,\vec{r}_n), \tag{1.88}$$

where $b_k(t)$ are time-dependent coefficients. Inserting this ansatz into the Equation (1.85)

$$\mathrm{i}\hbar\frac{\partial}{\partial t}\sum_k b_k(t)\phi_k = \hat{H}\sum_k b_k(t)\phi_k, \tag{1.89}$$

multiplying from the left with ϕ_j^* and integrating over all coordinates $\vec{r}_i$ one obtains

$$\mathrm{i}\hbar\frac{\mathrm{d}b_j(t)}{\mathrm{d}t} = \sum_k H_{jk}b_k(t). \tag{1.90}$$

To obtain this result, we have used the following properties:

1. the orthonormality of the molecular eigenfunctions

$$\int ... \int\int \phi_j^*\phi_k \mathrm{d}\vec{r}_1 ... \mathrm{d}\vec{r}_n = \langle\phi_j|\phi_k\rangle = \Delta_{jk} \tag{1.91}$$

2. the notation for matrix elements of the Hamiltonian

$$H_{jk} = \langle\phi_j|\hat{H}|\phi_k\rangle = \int ... \int\int \phi_j^*\hat{H}\phi_k \mathrm{d}\vec{r}_1 ... \mathrm{d}\vec{r}_n. \tag{1.92}$$

This system of equations can be written as a matrix equation

$$\mathrm{i}\hbar\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{b}(t) = \boldsymbol{H}\boldsymbol{b}(t) \tag{1.93}$$

where $\boldsymbol{b}(t)$=$(b_1, b_2, ..., b_N)^{\mathrm{T}}$ is a vector of coefficients and $\boldsymbol{H}$=$\{H_{jk}\}$ is the matrix representation of the Hamiltonian $\hat{H}$.

We now discuss the special case of **Rabi oscillations in a strongly driven two-level system**. We again consider the two-level system described previously, which now interacts with a coherent electromagnetic field $E = E_0\cos(\omega t + \phi)$. In this case, the quantities in Equation (1.93) are $H_{11} = \langle\phi_1|\hat{H}_{\mathrm{mol}}|\phi_1\rangle = E_1 = \hbar\omega_1$ and $H_{22} = E_2 = \hbar\omega_2$. The coupling matrix element for electric dipole coupling is

$$H_{12} = \langle\phi_1|\hat{V}_{\mathrm{el}}(t)|\phi_2\rangle \tag{1.94}$$

and we define

$$V_{12} = -\frac{\langle\phi_1|\hat{\vec{\mu}}_{\mathrm{z}}|\phi_2\rangle}{\hbar}E_0^{(z)}. \tag{1.95}$$

Since $\hat{\vec{\mu}}_{\mathrm{z}}$ is real, the case of real ϕ_1 and ϕ_2 leads to $V_{12} = V_{21} = V$. The time-dependent coefficients of the total wave function $\Psi(t) = b_1(t)\phi_1 + b_2(t)\phi_2$ must fulfill the system of coupled differential equations

$$\mathrm{i}\frac{\mathrm{d}b_1}{\mathrm{d}t} = \omega_1 b_1 + V\cos(\omega t + \phi)b_2 \tag{1.96}$$

$$\mathrm{i}\frac{\mathrm{d}b_2}{\mathrm{d}t} = V\cos(\omega t + \phi)b_1 + \omega_2 b_2. \tag{1.97}$$

This system of coupled differential equations has no simple closed-form solution but an approximate solution has been given by Rabi (*Phys. Rev.* **51**, 652 (1937)).

For the case of **near-resonant excitation** with the detuning $\Delta = (\omega_2 - \omega_1) - \omega = \omega_{12} - \omega$ the population of state 2 is

$$p_2(t) = |b_2(t)|^2 = \frac{V^2}{V^2 + \Delta^2}\left[\sin\left(\frac{t}{2}\sqrt{V^2 + \Delta^2}\right)\right]^2 = 1 - p_1(t), \tag{1.98}$$

i. e. the population undergoes periodic oscillations with the period

$$T = \frac{2\pi}{\sqrt{V^2 + \Delta^2}}. \tag{1.99}$$

Two cases of Rabi oscillations are illustrated in Figure 1.10. On resonance, *i. e.* for $\Delta = 0$ and $\omega = \omega_2 - \omega_1$, $T = 2\pi/V$. In this case $p_2(t)$ oscillates between 0 and 1. Off resonance, the

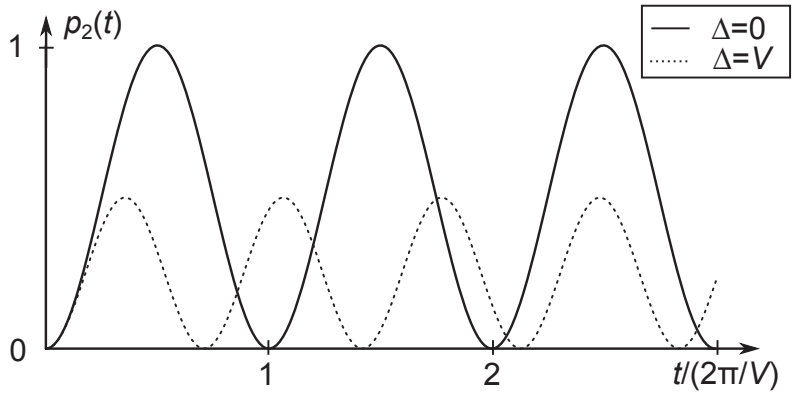


Figure 1.10: Temporal evolution of the excited state population $p_2(t)$ of a two-level system excited by radiation of circular frequency ω, detuned from resonance by $\Delta = \omega_{12} - \omega$. The time is given in units of the Rabi period $2\pi/V$.

oscillation is *faster* and $p_2(t)$ never reaches 1. The time-averaged excitation fraction amounts to

$$\langle p_2(\omega)\rangle_t = \frac{1}{2}\frac{V^2}{V^2+\Delta^2}, \tag{1.100}$$

i. e. a Lorentz function with FWHM (Full Width at Half Maximum) of $2V$ and also corresponds to the mean absorbed energy for excitation at circular frequency ω. Since $V^2 \propto E_0^2 \propto I_E$, this behavior is also known as intensity- or power broadening of spectral transitions that is frequently observed in laser-based spectroscopic experiments.

1.8 Time- and frequency-resolved spectroscopies

Two fundamentally different but complementary approaches to spectroscopy exist. The first and most widely used is *frequency*-domain spectroscopy. It relies on *tunable* sources of radiation that are designed to be as *narrow* as possible in frequency. These sources are used to measure energy intervals in atoms or molecules to the highest possible precision as illustrated in Figure 1.11a. The measured energy intervals are then used to construct or refine the Hamiltonian $\hat{H}_{\text{mol}}$ of the system. Once $\hat{H}_{\text{mol}}$ is known to the desired level of accuracy, all properties of the system, including time-dependent properties can be predicted. However, the complete determination of $\hat{H}_{\text{mol}}$ from spectroscopic measurements becomes increasingly difficult for larger molecules.

Time-domain spectroscopy relies on *ultrashort* laser pulses which, by virtue of the uncertainty relation, have a very *broad* spectrum. The large coherent bandwidth of the excitation pulse

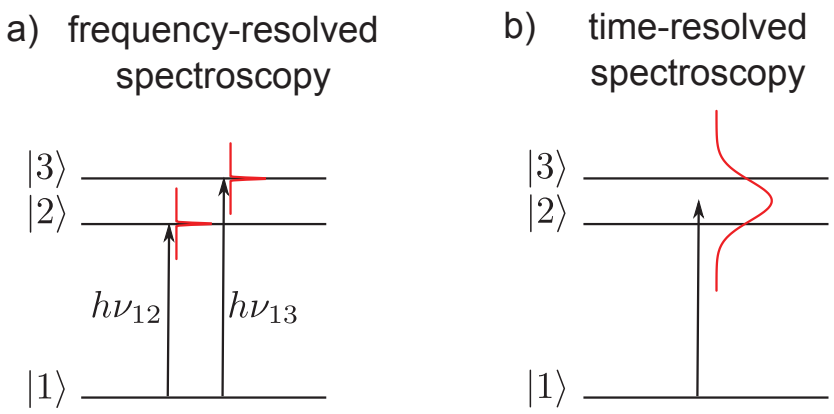


Figure 1.11: Principles of frequency-domain (a) and time-domain (b) spectroscopies. In frequency-domain spectroscopy, a narrow-band light source is tuned across the transitions of the molecule under study to determine the energy intervals. In time-domain spectroscopy, a short pulse with a broad spectrum is used to create a coherent superposition of several levels and the ensuing dynamics is studied.

is used to prepare a *coherent superposition* of molecular eigenstates whose *time evolution* is studied (see Figure 1.11b). This evolution provides information on $\hat{H}_{\text{mol}}$ that is complementary to the frequency-domain approach and may be simpler to interpret. An important example is the analysis of the mechanism of photostability of DNA which has been studied with time-domain measurements (see *e. g.* H. Satzger *et al.*, *Proc. Nat. Acad. Sci. USA*, **103** 10196 (2006)).

In the following sections, two simple quantum mechanical systems will be discussed to illustrate frequency-domain and time-domain measurements.

1.8.1 The harmonic oscillator

REMINDER: we consider a one-dimensional quantum mechanical system of two masses m_1 and m_2 held together by a potential $V(r) = \frac{1}{2}kr^2$ that is a quadratic function of their distance r. The Hamiltonian of this system can be written

$$\hat{H} = -\frac{\hbar^2}{2\mu}\frac{\partial^2}{\partial r^2} + \frac{1}{2}kr^2 \tag{1.101}$$

where $\mu = m_1 m_2/(m_1 + m_2)$ is the reduced mass of the system and k is a force constant

characterizing the potential. The eigenvalues and eigenfunctions of this Hamiltonian are

$$\frac{E_v}{hc} = \omega_e\left(v+\frac{1}{2}\right) \tag{1.102}$$

$$\phi_v(r) = \frac{1}{\sqrt{\sqrt{\pi}v!2^v}}H_v(r)e^{-\frac{1}{2}r^2}, \tag{1.103}$$

where $v = 0, 1, 2, ...$ is the vibrational quantum number, ω_e is the vibrational frequency expressed in cm^{-1} and $H_v(r)$ are the Hermite polynomials

$$H_0(r) = 1 \tag{1.104}$$

$$H_1(r) = 2r \tag{1.105}$$

$$H_2(r) = 4r^2 - 2 \tag{1.106}$$

etc.

Frequency-resolved spectroscopy measures the energy intervals between E_0 and E_1, E_1 and E_2 etc. Pure vibrational transitions in a harmonic oscillator can only be observed if $\Delta v = \pm 1$ and if the molecular dipole moment varies with r. In time-resolved spectroscopy, one can use an ultrashort laser pulse to prepare a *coherent superposition* of states ϕ_0 and ϕ_1 (through stimulated Raman scattering) and study its time evolution. We consider the coherent superposition, which is also called a **wave packet**,

$$\begin{aligned}\Psi(t) &= \frac{1}{\sqrt{2}}\left[\phi_0 e^{-\frac{iE_0t}{\hbar}} + \phi_1 e^{-\frac{iE_1t}{\hbar}}\right] \\ &= \frac{1}{\sqrt{2}}e^{-\frac{iE_0t}{\hbar}}\left[\phi_0 + \phi_1 e^{-\frac{i(E_1-E_0)t}{\hbar}}\right].\end{aligned} \tag{1.107}$$

For $t_0 = 0$, $|\Psi(t_0=0)|^2 = \frac{1}{2}(\phi_0+\phi_1)^2$.
For $t_1 = \frac{\hbar\pi}{(E_1-E_0)}$, *i. e.* $\frac{(E_1-E_0)t_1}{\hbar} = \pi$, $|\Psi(t_1)|^2 = \frac{1}{2}(\phi_0-\phi_1)^2$.
For $t_2 = \frac{2\pi\hbar}{(E_1-E_0)}$, $|\Psi(t_2)|^2 = |\Psi(t_0)|^2$.
The period of oscillation of $\Psi(t)$ is thus

$$T = \frac{h}{\Delta E}, \tag{1.108}$$

which is a universal relation valid for all wave packets consisting of two levels.
When many levels of the harmonic oscillator are coherently populated, the wave packet still undergoes a periodic motion. In the case of strong *resonant* excitation with monochromatic light $E = E_0\cos(\omega t)$, the time evolution of the wave packet is given by

$$|\Psi(q,t)|^2 = \frac{1}{\sqrt{\pi}}e^{-(q-q_0(t))^2} \tag{1.109}$$

as shown in the work of Marquardt and Quack (*J. Chem. Phys.* **90**, 6320 (1989)) where $q_0(t)$ represents the center of the wave packet and is given by

$$q_0(t) = -\frac{V_{01}}{\sqrt{2}}\left(\frac{\sin(\omega t)}{\omega}\right), \tag{1.110}$$

where $V_{01} = -\frac{\mu_{01}}{\hbar}E_0$.

Equation (1.109) shows that a quantum mechanical wave packet consisting of a large number of levels can behave in an essentially classical manner. This fact shows how classical and quantum mechanical systems are related.

1.8.2 The rigid rotor

REMINDER: the Hamiltonian for a linear rigid rotor in three dimensions is

$$\hat{H}_{\text{rot}} = \frac{\hat{J}^2}{2I}, \tag{1.111}$$

where $\hat{J}$ is the angular momentum operator (see Lecture Physical Chemistry III) and I is the moment of inertia of the rotor. The eigenfunctions $\Psi_{J,M}$ fulfill the Schrödinger equation

$$\begin{aligned} \hat{H}_{\text{rot}}\,\Psi_{J,M} &= E_J\,\Psi_{J,M} \\ \frac{1}{2I}\hat{J}^2\,\Psi_{J,M} &= \frac{\hbar^2}{2I}J(J+1)\,\Psi_{J,M} \end{aligned} \tag{1.112}$$

with the eigenvalues $E_J = \frac{\hbar^2}{2I}J(J+1) = hcBJ(J+1)$, where $B = \frac{\hbar^2}{2hcI}$ is the rotational constant of the rotor in cm^{-1}.

A frequency domain measurement measures the interval $E_{J'} - E_{J''}$ between the initial state J'' and the final state J'. We will show in later chapters that pure rotational transitions can only be observed if $\Delta J = J' - J'' = 0, \pm 1$ and if the molecule possesses a *permanent dipole moment*. In a time-domain measurement, one could use a linearly polarized ultrashort intense laser pulse to prepare a coherent superposition of rotational levels (through repeated stimulated Raman scattering). Each level $|J'', M''\rangle$ that was populated before the interaction with the laser pulse contributes a subset of coherently populated levels $|J', M'\rangle$

$$\Psi = \sum_{J'} a_{J',J''}|J', M' = M''\rangle, \tag{1.113}$$

where the $a_{J',J''}$ are obtained by solving the time-dependent Schrödinger equation for the interaction of the laser field with the molecule. A commonly used observable is the "degree

of axis alignment" $\langle \cos^2\theta \rangle$ given by

$$\langle \cos^2\theta \rangle(t) = \sum_{J''} P(J'') \sum_{J,J'} a^*_{J,J''}(t) a_{J',J''}(t) \langle J, M | \cos^2\theta | J', M' \rangle, \tag{1.114}$$

where $P(J'')$ is the initial thermal population of the rotational levels and the inner sum runs over all levels populated by the ultrashort laser pulse.

Figure 1.12 shows $\langle \cos^2\theta \rangle(t)$ for N_2 molecules exposed to a 100-fs laser pulse. The interaction with the laser pulse induces a "prompt alignment" followed by a fast decay towards near-random alignment (random alignment is characterized by $\langle \cos^2(\theta) \rangle = 1/3$) and revivals at every *half* of the classical rotation period $T = \frac{h}{\Delta E} = \frac{h}{hc(2B)} = \frac{1}{2cB} = 8.3\,\text{ps}$.

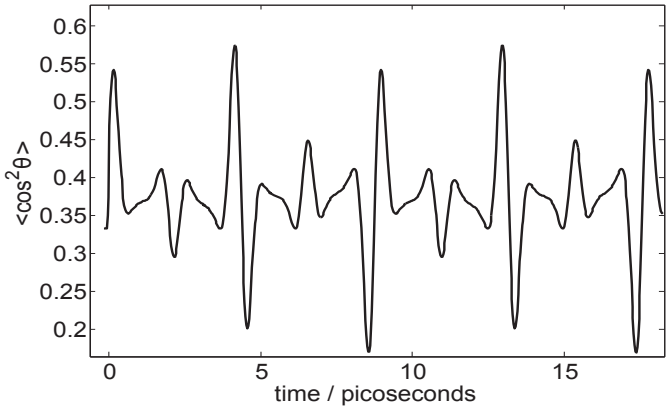


Figure 1.12: Evolution of the degree of axis alignment $\langle \cos^2\theta \rangle$ of N_2 molecules with an initial rotation temperature of $T = 30\,\text{K}$ that have been exposed to a 100 fs pulse of intensity $3{\times}10^{13}\,\text{W/cm}^2$.